



AI-BASED PROJECT TOPIC OFFERING

Generative AI (LLM) Topics for Project-Based Learning

Prepared for
Faculty of Computer Science
President University

Prepared by
Galih Permadi · LapisAI

June 2026 · Version 1.0

About This Offering

This document offers five industry-grounded project topics for President University students to complete for academic credit as part of Project-Based Learning. Each topic is drawn from a real problem that Xquisite AI and LapisAI solve for enterprise clients, so students graduate with portfolio work that mirrors actual industry practice.

Every project is built entirely on Generative AI (Large Language Models). Students learn to apply, orchestrate, and evaluate pre-trained LLMs — prompt engineering, retrieval-augmented generation (RAG), and AI agents — rather than training classical machine-learning models from scratch.

Each topic below includes its background, scope and deliverables, the skills students will build, step-by-step work guidance, the structure of the dummy data we provide, and a recommended technology stack. Teams can pick one topic; scope can be tuned with us at kickoff.

Project 1: Enterprise Knowledge Assistant (RAG)

Background

Large organizations hold thousands of internal documents — SOPs, policies, contracts, technical guides, and reports — scattered across drives and systems. Employees spend an estimated 20% of their working time just looking for the right information, and traditional keyword search often fails when people phrase a question differently from how the document is written.

Xquisite AI wants students to build a Retrieval-Augmented Generation (RAG) assistant that lets staff ask questions in natural language and receive accurate answers, each backed by a citation to the exact source document and page.

Scope & Deliverables

- A RAG pipeline that ingests and indexes PDF, DOCX, and TXT documents
- A chat interface for natural-language question answering over the document collection
- A citation feature: every answer shows the source document name and page
- An admin panel to upload new documents and review query logs
- A retrieval-quality evaluation (precision@k, recall@k, MRR, or the RAGAS framework)
- A simple business case for commercializing an enterprise-search product

Skills Students Will Build

Skill area	What students practice
Prompt Engineering	Grounding answers in retrieved context, enforcing citation behaviour, controlling hallucination
RAG Architecture	Chunking strategy, embeddings, semantic + keyword retrieval, re-ranking
Vector Databases	Storing and querying embeddings (ChromaDB / FAISS)
LLM Evaluation	Building ground-truth Q&A sets and measuring retrieval and answer quality
App Development	Shipping a usable chat UI and admin panel
Entrepreneurship	Business model and pricing for an enterprise-search tool

Work Steps

1. **Document Ingestion Pipeline** — Parse PDF/DOCX/TXT (PyMuPDF, python-docx). Chunk text into 500–1000 token segments with ~100 token overlap.
2. **Embedding & Vector Store** — Generate embeddings (sentence-transformers or an embeddings API) and store them in ChromaDB or FAISS.
3. **Retrieval** — Implement hybrid search (semantic + BM25 keyword) and re-rank the top candidates.
4. **Generation** — Pass retrieved context to an LLM and require answers that quote and cite their sources.
5. **UI Development** — Build a chat interface (Streamlit / Gradio) that displays answers, sources, and a confidence indicator.
6. **Evaluation** — Create ground-truth Q&A pairs and measure faithfulness and answer relevance (e.g. RAGAS).

Dummy Data Provided

A collection of ~50 simulated company documents (≈200–300 pages) plus a ground-truth Q&A set:

10 SOPs, 10 company policies, 10 operational reports, 10 technical documents, and 10 FAQ / knowledge-base articles. Students may also generate their own dummy documents from provided templates.

Question	Expected answer	Source document
How long is the probation period for new staff?	3 months, with an evaluation in week 12	SOP_Onboarding.pdf, p.5
How do I reset my password?	Raise a ticket to IT Helpdesk via the portal; ~1×24h	FAQ_IT_Support.docx
What is the work-from-home policy?	Max 2 days/week with manager approval	Policy_WFH.pdf, p.2

Recommended Tools & Technologies

- LangChain or LlamaIndex as the RAG framework
- ChromaDB, FAISS, or Weaviate as the vector store
- HuggingFace sentence-transformers for embeddings
- An LLM via API (OpenAI / Anthropic) or a local model (Llama 3 via Ollama)
- Streamlit or Gradio for the chat interface
- RAGAS for evaluation

Project 2: AI Customer Support Copilot

Background

B2B companies receive thousands of support tickets every month across email, web forms, and chat. Triageing each ticket to the right team and writing a reply is done manually today, which is slow and leads to mis-routing of up to 25%.

Xquisite AI wants students to build an LLM-powered copilot that triages incoming tickets and drafts a context-aware reply for a human agent to review. Important: this project uses pre-trained large language models with prompt engineering and retrieval — no machine-learning model is trained from scratch.

Scope & Deliverables

- Ticket triage that classifies category and priority using few-shot prompting (no model training)
- An auto-drafted reply grounded in a help-centre knowledge base (RAG)
- Confidence scoring and escalation logic that routes low-confidence or sensitive tickets to a human
- A human-in-the-loop review screen where the agent edits and approves before sending
- A monitoring dashboard: ticket volume by category, draft-acceptance rate, average handling time
- An evaluation: classification accuracy versus labels and LLM-as-judge scoring of draft quality

Skills Students Will Build

Skill area	What students practice
Prompt Engineering	Few-shot classification, structured (JSON) output, tone-controlled drafting
RAG	Grounding replies in approved help-centre content
Structured Outputs	Reliable JSON / function-calling for downstream automation
Human-in-the-Loop Design	Confidence thresholds, escalation, agent review flow
LLM Evaluation	LLM-as-judge rubrics and accuracy measurement
Entrepreneurship	ROI case for support automation

Work Steps

1. **Data Exploration** — Review the ticket dataset: category distribution, channel mix, typical length and tone.
2. **Triage Prompt Design** — Few-shot prompt returning JSON (category, priority, sentiment). No training — pure prompting.
3. **Knowledge Base + RAG** — Index help-centre articles so replies are grounded in approved answers.
4. **Reply Drafting** — Prompt the LLM to draft a policy-safe, on-tone reply that cites the relevant article.
5. **Escalation Logic** — Define confidence thresholds; route uncertain or sensitive cases to humans.

6. **Dashboard & Review UI** — Build the monitoring dashboard and the agent edit-and-approve screen.

7. **Evaluation** — Measure classification accuracy vs labels; score drafts with an LLM-as-judge rubric.

Dummy Data Provided

tickets_dataset.csv (~3,000 rows) of simulated support tickets, plus 20–30 help-centre articles used to ground replies.

ticket_id	subject	channel	true_category
TK-0001	Cannot log in to the dashboard since yesterday	email	Technical
TK-0002	December invoice not received yet	form	Billing
TK-0003	Request to add a new user to our account	chat	Account

Additional columns: body (long text), customer_id, customer_tier, timestamp, true_priority.

Recommended Tools & Technologies

- An LLM via API (OpenAI / Anthropic) or a local model
- LangChain for orchestration
- pydantic for structured outputs
- ChromaDB for the knowledge base
- Streamlit or React + Chart.js for the dashboard and review UI

Project 3: Conversational Data Analyst (Chat with Your Data)

Background

Business users in sales, operations, and finance constantly need answers from company databases, but most cannot write SQL — so they wait in a queue for an analyst. Every simple question becomes a ticket.

Xquisite AI wants students to build an LLM agent that turns a natural-language question into a safe SQL query, runs it against a sample database, and returns the answer with an automatic chart and a plain-language explanation.

Scope & Deliverables

- A natural-language-to-SQL agent that is schema-aware and read-only (no data can be modified)
- Query execution against a sample database, returning the answer, an auto-selected chart, and an explanation
- Handling of ambiguous questions through clarifying questions and multi-turn follow-ups
- A query log and a “show the SQL” transparency view
- An evaluation on a benchmark question set (execution accuracy)
- A business case for self-service analytics

Skills Students Will Build

Skill area	What students practice
Prompt Engineering	Schema-aware prompting and constraint setting
AI Agents & Tool Use	Letting an LLM call a database tool and react to results
Guardrails & Safety	Blocking write/DDL queries, validating generated SQL
SQL & Data Modelling	Understanding schemas, joins, and aggregations
Data Visualization	Auto-selecting an appropriate chart for the result
Entrepreneurship	Value case for democratized data access

Work Steps

1. **Set Up the Sample Database** — Load a small retail schema (customers, orders, order_items, products, payments) from seed CSVs.
2. **Schema-Aware Prompting** — Inject table schemas and sample rows so the LLM understands the data.
3. **SQL Generation + Validation** — Generate SQL and validate it is read-only; reject any write/DDL statements.
4. **Execute & Visualize** — Run the query, format results, and auto-select a suitable chart.
5. **Explanation Layer** — Have the LLM summarize the result in business language.
6. **Guardrails & Clarifications** — Ask clarifying questions for vague requests; log every query.
7. **Evaluation** — Run 50+ benchmark questions with gold answers and measure execution accuracy.

Dummy Data Provided

A SQLite database seeded from CSVs ($\approx$ 1,000 customers, 10,000 orders, 200 products, plus payments) and a benchmark set of natural-language questions with expected answers.

Natural-language question	Expected answer (concept)	Tables used
Which 5 products sold the most last month?	Top products by quantity in the period	orders, order_items, products
What is our total revenue this quarter?	Sum of paid order amounts in the quarter	orders, payments
How many new customers joined in May?	Count of customers created in May	customers

Recommended Tools & Technologies

- LangChain SQL agent or LlamaIndex
- SQLite or PostgreSQL
- An LLM via API (OpenAI / Anthropic)
- Plotly for charts
- Streamlit for the interface

Project 4: Intelligent Document Processing (Extraction & Automation)

Background

Finance and operations teams spend hours manually keying data from invoices, purchase orders, receipts, and forms into their systems. The work is slow, costly, and error-prone — and it scales badly as volume grows.

Xquisite AI wants students to build an intelligent document-processing pipeline that uses a multimodal LLM to read a document and output clean, structured data, with validation and a human review step before it is exported.

Scope & Deliverables

- Upload of a document (PDF or image) and LLM extraction of structured fields per document type (invoice, PO, receipt)
- Validation rules (totals reconcile, required fields present, format checks) that flag issues
- A human-in-the-loop review screen to correct and approve extracted data
- Export to CSV/JSON and a mock API endpoint
- A field-level accuracy evaluation against ground-truth labels
- A cost-benefit business case versus manual data entry

Skills Students Will Build

Skill area	What students practice
Prompt Engineering	Instructing the model to return exactly the fields needed
Multimodal LLMs	Reading scanned documents and images (vision)
Structured Outputs	Schema-constrained extraction (pydantic / JSON schema, function calling)
Validation Logic	Business-rule checks and confidence flagging
Human-in-the-Loop Design	Review, correct, approve workflow
Entrepreneurship	Cost-benefit and automation ROI

Work Steps

1. **Define Extraction Schemas** — Model each document type as a structured schema (pydantic), e.g. vendor, date, line items, totals.
2. **Document Loading** — Convert PDFs/images to a model-readable form (PyMuPDF / pdf2image).
3. **Extraction** — Prompt a multimodal LLM to return structured output; use a vision model for scans.
4. **Validation & Flagging** — Apply business rules (e.g. line items must sum to the total) and flag low-confidence fields.
5. **Review UI** — Show the document next to the extracted fields for edit-and-approve.
6. **Export & API** — Export approved data to CSV/JSON and a mock API endpoint.
7. **Evaluation** — Measure field-level precision/recall against the labelled ground truth.

Dummy Data Provided

≈200 synthetic documents (invoices, purchase orders, and receipts) as PDF/PNG, plus `ground_truth.csv` with the correct labelled fields for evaluation.

doc_id	doc_type	vendor	total_amount
DOC-001	invoice	PT Sumber Makmur	12,450,000
DOC-002	purchase_order	CV Mitra Teknik	3,900,000
DOC-003	receipt	Toko Sentosa	275,000

Additional columns: invoice_date, due_date, line_items (JSON), tax_amount, currency.

Recommended Tools & Technologies

- A multimodal LLM: OpenAI GPT-4o (vision) or a vision-capable local model via Ollama
- PyMuPDF or pdf2image for document loading
- pydantic for schema-constrained extraction
- Streamlit for the review UI
- JSON schema / function calling for structured output

Project 5: AI Content Studio & Brand-Voice Assistant

Background

Marketing teams must produce a high volume of on-brand content across many channels — social posts, email, blog — and do it fast. Writing everything manually does not scale, while generic AI output sounds off-brand. Teams also need to understand how the public reacts to what they publish.

Xquisite AI wants students to build a Generative-AI content studio that writes on-brand content from a brief, repurposes one asset into many formats, and produces an LLM-based sentiment digest of public comments. The sentiment digest uses zero-shot LLM classification and summarization — no machine-learning model is trained.

Scope & Deliverables

- A brand-voice profile (system prompt / few-shot from sample past posts) the generator must follow
- Multi-format generation from a single brief (e.g. blog → LinkedIn post, Instagram caption, email, X thread)
- A content-calendar suggester
- An LLM-based audience-sentiment and theme digest over public comments (zero-shot, no training)
- A simple approval/edit UI and export
- A market-sizing and business case for an AI content tool

Skills Students Will Build

Skill area	What students practice
Prompt Engineering	Persona / brand-voice prompts and few-shot conditioning
Content Repurposing	Transforming one asset into channel-specific formats
Zero-shot LLM Tasks	Classification and summarization of comments without training
Evaluation	Brand-fit rubric and LLM-as-judge scoring
UX for AI	Approval and editing workflow for generated content
Entrepreneurship	Market sizing for social-content tooling in Indonesia

Work Steps

1. **Build the Brand-Voice Profile** — Derive a voice profile from brand guidelines and sample past posts.
2. **Generation Engine** — From a brief, generate a draft that respects the voice and format constraints.
3. **Repurposing** — Turn one approved asset into multiple channel formats automatically.
4. **Sentiment & Theme Digest** — Zero-shot LLM classification of comments into sentiment and themes; produce a weekly summary and flag negative spikes with a simple threshold.
5. **Calendar & Approval UI** — Suggest a posting calendar; provide an approve/edit screen.
6. **Evaluation** — Score outputs against a brand-fit rubric (LLM-as-judge) with human spot-checks.

Dummy Data Provided

brand_guidelines (a short brand voice document) and sample_posts.csv of past on-brand posts to learn the voice, plus comments_dataset.csv (~5,000 rows) for the sentiment digest.

post_id	comment text	channel	sentiment
P-0001	The new feature is super helpful, love the auto-save!	instagram	positive
P-0002	App keeps crashing after the update, please fix	twitter	negative
P-0003	Anyone know how to claim the promo?	instagram	neutral

Additional columns: timestamp, likes, shares, brand_mentioned, follower_count.

Recommended Tools & Technologies

- An LLM via API (OpenAI / Anthropic) or a local model
- LangChain for orchestration
- Streamlit for the studio interface
- Optional image generation (DALL·E / Stable Diffusion) for visual posts

Collaboration & Contact

About Xquisite AI

Xquisite AI is a B2B AI and Data Analytics consultancy that has operated since 2016. With experience serving 90+ enterprise clients and delivering 400+ projects, we have deep expertise implementing AI solutions across industries. LapisAI is our Generative-AI product arm; the topics in this document reflect real LLM use cases we deliver for industry.

What We Provide

- A complete dummy dataset for each chosen project, delivered within one week of project kickoff
- A short kickoff briefing to align scope with each team
- Mentoring sessions as needed (suggested bi-weekly check-ins)
- Evaluation and feedback on the final presentations

Indicative Timeline

Relative to a standard semester — to be aligned with President University's academic calendar.

Phase	Period	Activities
Initiation	Weeks 1–2	Team formation, detailed briefing, dataset distribution
Development	Weeks 3–12	Build, bi-weekly mentoring, milestone reviews
Final Presentation	Weeks 13–14	Product demo, evaluation, feedback
Showcase	End of term	Exhibition of the best projects

Contact

LapisAI · Xquisite AI

Email: galih@xquisite.ai